

Transaction

Project Name: CRD
Group No: 10
Members: [names]
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> describes a local unit of database processing.

Transaction Processing System

> System with large databases and hundreds of concurrent users.

Transaction Processing

1. Single user DBMS

at most one user can access the system

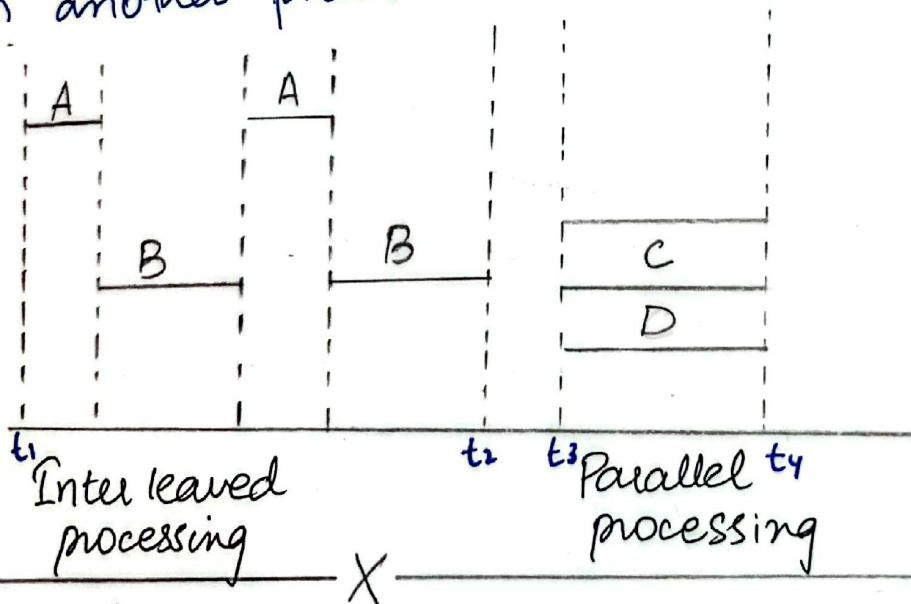
2. Multiuser DBMS

many user can access the system (database) concurrently.

3. Multi Programming

> allows OS to execute multiple processes concurrently.

> Executes command from one process, then suspend that process and executes commands from another process.



> Transaction

- set of logically related operation contains a group of tasks.
- transaction represents a change in database
- It is an action or a series of action
- performed by a single user to perform operations for accessing database

Example

Suppose an employee of bank transfer Rs 800 from X's account to Y's account.

→ X's Account

Open-Account (X) = 2000

Old-Balance = X-Balance = 2000

New-Balance = Old-Balance - 800 = 2000 - 800 = 1200

X-Balance = New-Balance = 1200

close-Account (X) = 1200

→ Y's Account

Open-Account (Y) = 2000

Old-Balance = Y-Balance = 2000

New-Balance = Old-Balance + 800 = 2000 + 800 = 2800.

Y-Balance = New-Balance = 2800

close-Account (Y) = 2800.

- > A transaction represents a complete and meaningful unit of work in a database
- > All the operations within a transaction are logically connected and must be executed completely that is if one fails, the entire transaction system fails

Begin/End

- > Every transaction has clear boundaries defined using begin(commit) and end(roll back).
- > It helps database understand where a transaction starts and where it ends
- > This ensures that either all operations succeed together or none are applied